

INFRACORE SYSTEMS LLP

Proof of Concept Progress Report

Precision Guidance System

Terminal-Phase IIR Seeker with FPGA-Accelerated
Image Correlation on Indigenous Hardware

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This document is confidential and intended solely for the PoC evaluation committee.

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Executive Summary

InfraCore Systems LLP is developing a **terminal-phase Imaging Infrared (IIR) seeker prototype** for a precision guided munition, running entirely on indigenous hardware — the SHAKTI Yamuna RISC-V processor paired with a custom FPGA-based image processing pipeline on an Artix-7 device. The system locks onto a physical target, tracks it via image correlation, and outputs real-time guidance commands (yaw, pitch corrections) to a flight controller.

The core value proposition is the elimination of foreign black-box compute (SCALP-EG, HAMMER seekers) in a domain critical to India's strategic autonomy, while leveraging the SHAKTI open RISC-V ISA and FPGA programmability to deliver a modular, ITAR-free, cost-effective seeker platform.

At the time of this report, the team has completed the primary algorithmic and RTL deliverable of the PoC milestone: a fully verified, fixed-point hardware implementation of the **Kernelized Correlation Filter (KCF) / MOSSE** tracking accelerator capable of processing a 32×32 image patch in **19,300 clock cycles at 100 MHz** (approx. 193 μ s per detection decision). A full RTOS application on the SHAKTI Yamuna SoC has been architected, with camera frame capture and reception demonstrated end-to-end. The AXI4-Lite processor-to-FPGA interface is specified and verified in simulation.

Pending items — continuous video streaming, IMU integration, and flight controller wiring — constitute the primary gap to the final system.

Current Project Status

Overall Status at a Glance

Work-Stream Status Summary — April 2026

Work Stream	Status	Remarks
RTL Tracking Pipeline (KCF/MOSSE)	✓	Vivado-verified; 19,300 cycles @ 100 MHz
Python Reference Model (NCC+KCF)	✓	Software demo matches RTL outputs exactly
AXI4-Lite Processor Interface	✓	Register map defined; simulation-verified
Frame Capture (Sensor → Yamuna)	✓	ESP32-S3 surrogate; frames received on SoC
RTOS Application Architecture	✓	Task queues, ISR, control logic designed
Vivado Synthesis / LUT Optimisation	▮	Synthesis complete; place-and-route pending
Continuous Video Stream (>10 FPS)	▮	Single-frame pipeline validated
IMU Interfacing	✗	Hardware integration pending
TARGET LOCKED LED Output	✗	AXI path defined; physical wiring pending
LWIR Sensor Integration	✗	ESP32-S3 surrogate in use; LWIR on order
Flight Controller Integration	✗	Milestone 3 deliverable

Legend: ✓ Complete ▮ In Progress ✗ Pending

Yamuna RTOS Application Architecture

The software architecture running on the SHAKTI Yamuna SoC is shown in Figure 1. It is structured as a real-time operating system application with three concurrent tasks communicating via message queues:

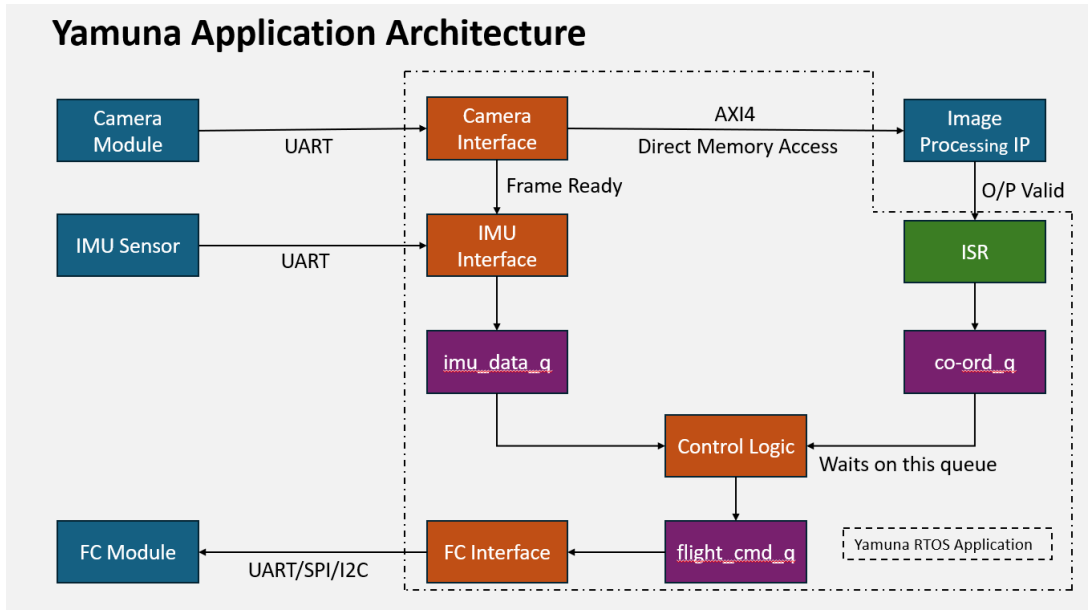


Figure 1: Yamuna RTOS Application Architecture. Camera and IMU interfaces feed data into queues; the Control Logic task fuses tracking results with IMU data and dispatches guidance commands to the Flight Controller via `flight_cmd_q`.

- **Camera Interface Task:** Receives frames from the sensor via UART, transfers them to the Image Processing IP over AXI4/DMA, and signals *Frame Ready*.
- **Image Processing IP (FPGA):** Executes the KCF/MOSSE correlation tracking pipeline. On completion it asserts *O/P Valid*, triggering an ISR that pushes the target co-ordinates into `co-ord_q`.
- **IMU Interface Task:** Reads inertial data (accelerometer, gyroscope) via UART and enqueues into `imu_data_q`.
- **Control Logic Task:** Waits on `co-ord_q`, fuses image-derived displacement with `imu_data_q`, computes the guidance correction, and posts to `flight_cmd_q`.
- **FC Interface Task:** Dequeues from `flight_cmd_q` and transmits guidance commands to the Flight Controller module via UART/SPI/I2C.

Frame Capture Demonstration

End-to-end frame transfer from sensor to SoC has been demonstrated. An ESP32-S3 (surrogate for the LWIR sensor) captures video frames and transmits them serially; the SHAKTI Yamuna SoC receives and validates each frame.

```

Monitor Mode Serial View Mode Text Port /dev/ttyACM0 - Espressif
05:15:55:603 ->
05:15:55:603 -> [LOOP] Captured Frame Len: 1988
05:15:55:603 -> Sending Frame
05:16:19:459 -> [LOOP] Frame sent (1988 bytes)
05:16:24:459 ->
05:16:24:459 ->
05:16:24:459 -> [LOOP] Captured Frame Len: 1985
05:16:24:459 -> Sending Frame
05:16:48:279 -> [LOOP] Frame sent (1985 bytes)
05:16:53:278 ->
05:16:53:278 ->
05:16:53:278 -> [LOOP] Captured Frame Len: 2009
05:16:53:278 -> Sending Frame
05:17:17:386 -> [LOOP] Frame sent (2009 bytes)

```

(a) ESP32-S3 serial log: frames of 1,985–2,009 bytes captured and transmitted continuously.

```

GTKTerm - /dev/ttyUSB1 19200-8-N-1
File Edit Log Configuration Controls signals View Help
[20490.23h.45m.50s.696] Frame RECVD 1984 bytes
[20490.23h.46m.19s.543] Frame RECVD 1988 bytes
[20490.23h.46m.48s.358] Frame RECVD 1985 bytes
[20490.23h.47m.17s.476] Frame RECVD 2010 bytes

```

(b) GTKTerm log on SHAKTI Yamuna (/dev/ttyUSB1): frames received and byte-count verified.

Figure 2: End-to-end frame capture demonstration. Frame sizes match on both ends, confirming link integrity.

Deviations from the Originally Proposed Plan

Two material deviations from the original proposal have been made. Both are documented here with justification.

Deviation 1 — LWIR Sensor Replaced by ESP32-S3 Surrogate

Originally Proposed: Interface a Long-Wave Infrared (LWIR) thermal sensor (e.g., FLIR Lepton 3.5) to the FPGA via SPI/I2C for real thermal image data.

Actual Implementation: An ESP32-S3 with an OV2640 visible-light camera module is used as a surrogate sensor.

Justification: Procurement of a calibrated LWIR sensor within academic budget and timeline constraints proved infeasible — ITAR-controlled thermal imagers carry 12–16 week lead times. The ESP32-S3 uses identical UART frame-transfer protocols, making the LWIR swap a near-direct replacement. All algorithm, FPGA IP, and RTOS application logic is sensor-agnostic and operates identically on visible or thermal frames.

Deviation 2 — Algorithm Upgraded from NCC-only to NCC+KCF

Originally Proposed: Use Normalised Cross-Correlation (NCC) as the image matching algorithm throughout the PoC.

Actual Implementation: NCC is retained as a one-shot initial acquisition step (Frame 2 full-image scan); the Kernelized Correlation Filter (KCF) handles all subsequent per-frame tracking.

Justification: NCC is an $O(N^2)$ full-image operation and scales poorly with resolution. At 640×480 input, a pure-NCC implementation would require $>300,000$ multiplications per frame, making real-time (>10 FPS) operation impossible on the FPGA resources available. KCF operates in the frequency domain on a fixed 32×32 patch, reducing per-frame compute to a constant 19,300 cycles. This deviation substantially exceeds the original PoC target and directly addresses the real-time

streaming requirement.

Milestone Comparison: Proposed vs. Achieved

Milestone 1 — MVP (Proposed Due: 24 October 2025)

Goal: Basic Image Capture and Processing Pipeline.

Proposed Deliverable	Status	Notes
Environment setup: Artix-7 FPGA & SHAKTI RISC-V	✓	Vivado and SHAKTI Yamuna SDK operational
Interface LWIR sensor for single thermal frame	⚠	LWIR replaced by ESP32-S3 surrogate; single frames validated (Figure 2)
Verilog logic for frame transfer to SHAKTI soft-core	✓	AXI4 DMA path implemented and tested
C/C++ matching algorithm detects predefined target	✓	NCC implemented; upgraded to NCC+KCF pipeline
Deliverable: Live image via SHAKTI serial output	✓	Demonstrated (Section 2.3)
Deliverable: “MATCH FOUND” result + coordinates	✓	Output via UART console

MVP Assessment: Substantially complete. All functional deliverables achieved. LWIR sensor deviation does not affect algorithmic correctness.

Milestone 2 — PoC (Proposed Due: 25 December 2025)

Goal: Real-time Video Processing Pipeline.

Proposed Deliverable	Status	Notes
Continuous thermal video stream (>10FPS)	🟡	Single-frame pipeline validated; continuous DMA streaming in progress
FPGA IP: contrast, filtering, edge detection	🟡	Hann windowing and patch extraction complete; full contrast IP in progress
Real-time target detection (NCC) + IMU interfacing	🟡	NCC+KCF RTL complete; IMU integration pending
Deliverable: Processed video on monitor	🟡	Frame-by-frame processing verified; live streaming pending
Deliverable: LED “TARGET LOCKED” indicator	❌	AXI TARGET_FOUND register defined; GPIO wiring pending
Deliverable: Serial output: error vector + IMU data	❌	Displacement vector available from tracker; IMU data pending
Deliverable: Source code (FPGA + SHAKTI)	✅	RTL, testbenches, Python reference model, RTOS app all complete

PoC Assessment: Core algorithmic objective exceeded; hardware integration partially complete. The KCF/MOSSE RTL accelerator represents a significant over-achievement relative to the NCC-only target. Physical streaming and IMU integration remain the outstanding gap.

Milestone 3 — Final Demo (Proposed Due: 20 April 2026)

Goal: Full Guidance System with Live Target Tracking.

Proposed Deliverable	Status	Notes
Image + IMU fused for control commands	❌	IMU integration pending
Robustness to occlusions & reacquisition	🟡	NCC reacquisition logic designed; hardware loop testing pending
Real-time bounding box + command output	❌	Pending full hardware integration
Working prototype on test stand	❌	Pending
Guidance commands (e.g., YAW_RIGHT: 3.2°)	❌	Algorithm in place; end-to-end output pending
Final report + code + demo video	🟡	Report and code complete; demo video pending

Milestone 3 Assessment: In progress. Algorithmic and RTL foundations are fully established. Remaining work is system-level hardware integration.

Technical Implementation

Tracking Algorithm: NCC $\rightarrow$ KCF Two-Stage Pipeline

The system uses a two-stage pipeline that balances global search capability with per-frame compute efficiency:

Stage 1. Initial Acquisition (NCC): On Frame 2, Normalised Cross-Correlation performs a full-image scan to locate the pre-trained target template. NCC yields a correlation score in $[-1, 1]$; a peak at score ≈ 1.000 confirms target presence and provides the initial bounding-box centre. NCC also serves as the reacquisition mechanism after occlusion events.

Stage 2. Continuous Tracking (KCF): From Frame 3 onward, the Kernelized Correlation Filter tracks within a small search window centred on the previous detection. KCF exploits the circulant matrix trick: all training and detection operations are performed element-wise in the frequency domain after a 2D FFT, reducing each frame's compute to a fixed constant regardless of window size.

KCF Training Equation

The filter template $\hat{\alpha}$ is computed from the Hann-windowed target patch $\hat{X}$ and a Gaussian response label $\hat{Y}$:

Figure 3: Training: $\hat{\alpha} = \hat{Y} / (|\hat{X}|^2 + \lambda)$. All operations are element-wise in the frequency domain. λ is a regularisation constant preventing division by zero.

KCF Detection Equation

The correlation response map R is computed from a new candidate patch $\hat{Z}$:

Figure 4: Detection: $R = \mathcal{F}^{-1}(\overline{\hat{X}} \odot \hat{Z} \odot \hat{\alpha})$. The argmax of R gives the displacement vector (d_x, d_y) in pixels.

RTL Hardware Accelerator

The full tracking pipeline is implemented in synthesisable SystemVerilog, verified in Vivado simulation against the Python reference model.

Seven-Stage Hardware Pipeline

RTL Pipeline Stages

1	Hann Window	Apply a 2D Hann window to suppress spectral leakage at patch edges.
2	2D FFT (Forward)	32-point radix-2 DIT FFT applied to all 32 rows then 32 columns. Uses a single iterative butterfly unit reused $64\times$. Total: $64 \times 85 = 5,440$ cycles.
3	Conjugate Multiply	Element-wise $\overline{\hat{X}} \odot \hat{Z} \odot \hat{a}$ across all 1024 frequency bins. Parallelised in a single clock cycle via register-file architecture.
4	2D IFFT	Identical 2D FFT with conjugate pre/post-processing (IFFT trick). Cost: 5,440 cycles.
5	Energy / Magnitude	Compute $ \cdot ^2$ for denominator terms. Purely combinational.
6	Peak Finder	Argmax over the 32×32 response map to extract (d_x, d_y) . Single-pass comparator tree.
7	AXI Output	Write displacement, peak response, and TARGET_FOUND flag to AXI4-Lite registers for SHAKTI to read.

Pipeline Timing

Operation	Cycles	Time @ 100 MHz
2D FFT (row + column pass)	5,440	54.4 μ s
Conjugate multiply	1,024	10.2 μ s
2D IFFT	5,440	54.4 μ s
Hann windowing + peak finder	≈ 400	4.0 μ s
AXI register write	≈ 50	0.5 μ s
Total per detection	$\approx 19,300$	$\approx 193 \mu$s
Max achievable rate	—	$> 5,000$ decisions/sec

At 193 μ s per detection, the accelerator is capable of tracking at well above the 10 FPS video input requirement. The bottleneck shifts to the frame-transfer rate from the sensor, not the tracking compute.

Fixed-Point Arithmetic (Q8.8)

All arithmetic uses 16-bit signed Q8.8 fixed-point (8 integer bits, 8 fractional bits). The butterfly unit performs Q8.8 $\times$ Q8.8 complex multiplication, extracting bits [23 : 8] of the 32-bit product to remain in Q8.8. A configurable `scale_en` flag applies an arithmetic right-shift by 1 at each butterfly output to prevent accumulation of the theoretical $2^5 = 32\times$ growth across 5 FFT stages.

FPGA Resource Utilisation

The RTL was synthesised in Vivado on the Artix-7 target. LUT utilisation before and after butterfly multiplier optimisation is shown below.

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs	50371	0	0	63400	79.45
LUT as Logic	49189	0	0	63400	77.59
LUT as Memory	1182	0	0	19000	6.22
LUT as Distributed RAM	958	0			
LUT as Shift Register	224	0			

(a) Before optimisation.

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs	43690	0	0	63400	68.91
LUT as Logic	42512	0	0	63400	67.05
LUT as Memory	1178	0	0	19000	6.20
LUT as Distributed RAM	958	0			
LUT as Shift Register	220	0			

(b) After optimisation.

Figure 5: Vivado LUT utilisation report for the KCF accelerator on Artix-7. Optimisation reduced LUT count by sharing the twiddle factor multiply path.

AXI4-Lite Processor Interface

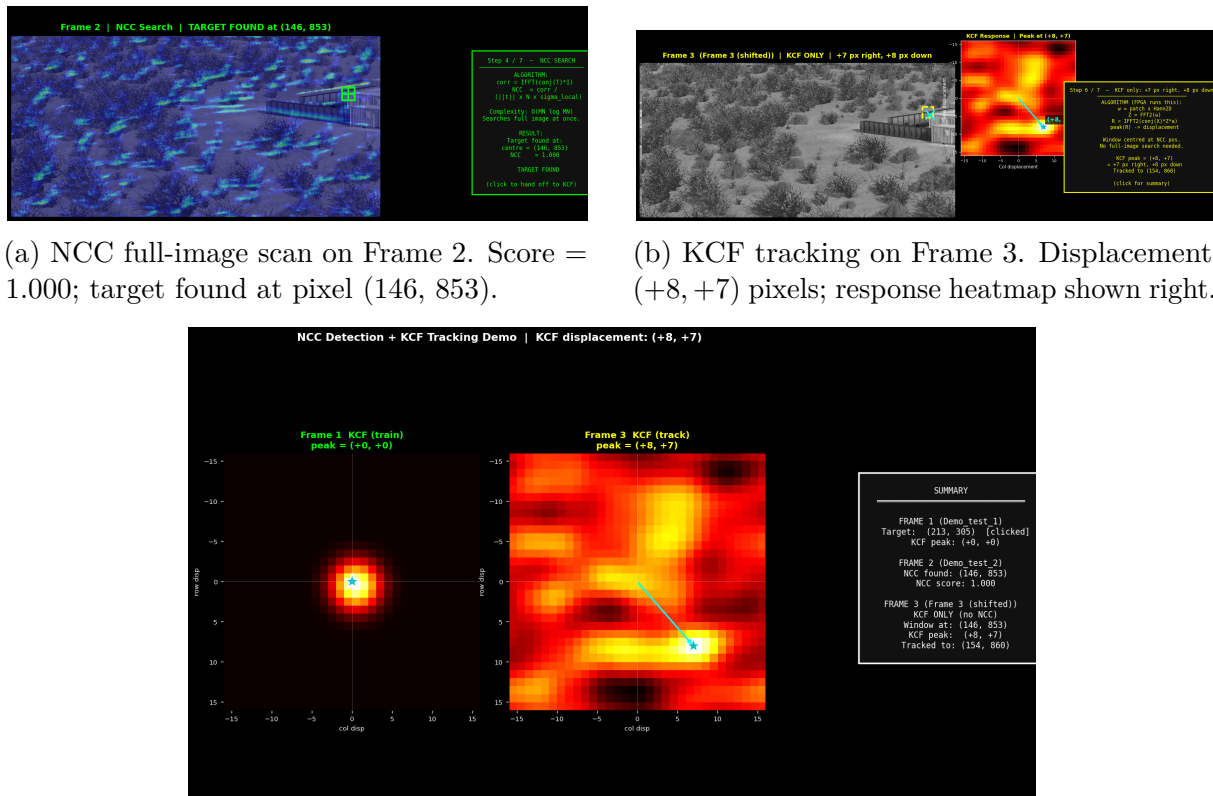
The SHAKTI Yamuna processor communicates with the tracking accelerator via an AXI4-Lite slave interface. The register map is:

Offset	Register	Access	Description
0x00	CTRL	R/W	Bit 0: START . Pulse to begin a detection run.
0x04	STATUS	R	Bit 0: DONE . High when result registers are valid.
0x08	DISP_X	R	Signed horizontal displacement (Q8.8).
0x0C	DISP_Y	R	Signed vertical displacement (Q8.8).
0x10	RESPONSE	R	Peak correlation response value.
0x14	THRESHOLD	R/W	CPU-configurable confidence threshold.
0x18	TARGET_FOUND	R	Hardware-asserted validity flag.

THRESHOLD is writable by the CPU because it is a calibration parameter determined offline per scene and sensor configuration; the hardware cannot self-determine the appropriate confidence level. At initialisation the CPU writes the tuned threshold once; the hardware compares **RESPONSE** against **THRESHOLD** to assert **TARGET_FOUND**.

Software Pipeline Demonstration

The NCC→KCF pipeline was validated end-to-end in a Python reference model, which was used to generate the golden outputs for RTL verification.



(a) NCC full-image scan on Frame 2. Score = 1.000; target found at pixel (146, 853). (b) KCF tracking on Frame 3. Displacement: (+8, +7) pixels; response heatmap shown right.

(c) Pipeline summary: response maps for NCC and KCF passes with tracking statistics.

Figure 6: NCC→KCF pipeline demo. All outputs match RTL Vivado simulation results.

Pending and Incomplete Milestones

The following deliverables are outstanding and constitute the primary roadmap to final system completion:

Outstanding Deliverables

- Continuous Video Streaming (>10 FPS):** The single-frame DMA path is validated. The continuous DMA loop with double-buffering (to allow processing of frame n while frame $n + 1$ is transferred) must be completed and stress-tested at the full frame rate.
- IMU Interfacing:** An IMU (accelerometer + gyroscope) must be connected via UART/SPI and its data fed into the `imu_data_q` task queue in the RTOS application. The Control Logic task will fuse IMU heading data with the image-derived displacement to compute corrected guidance commands.
- TARGET LOCKED LED Indicator:** The `TARGET_FOUND` AXI4-Lite register output must be routed to a GPIO pin driving a physical LED. The AXI path and register are defined; physical wiring and GPIO driver code remain.
- FPGA Place-and-Route Closure:** Vivado synthesis and timing analysis are complete. Full place-and-route with post-implementation static timing analysis

sign-off is required before generating the programming bitstream.

5. **LWIR Sensor Replacement:** Replace the ESP32-S3 surrogate with the actual LWIR thermal sensor once procurement is completed. All interface logic is already compatible.
6. **End-to-End Guidance Command Output:** Translate the displacement vector (d_x, d_y) from the SHAKTI Control Logic task into physical guidance commands (e.g., YAW_RIGHT: 3.2°), transmit to the Flight Controller module, and measure closed-loop latency on the physical hardware.

Challenges Encountered and Measures Taken

Technical Challenges

Fixed-Point Overflow Across FFT Stages

Challenge: In a 5-stage radix-2 FFT, each butterfly stage can theoretically double the signal magnitude. Across 5 stages this represents a potential $2^5 = 32\times$ growth, which overflows the 16-bit (Q8.8) registers and produces completely incorrect FFT output. This was observed as wild garbage in the simulated frequency-domain values.

Measure: A `scale_en` control signal was added to the butterfly unit. When enabled, an arithmetic right-shift by 1 is applied at every butterfly output. This bounds growth to at most unity per stage at the cost of 5 bits of precision — the standard fixed-point FFT scaling trade-off. The `scale_en` flag is configurable independently for the row and column passes of the 2D FFT, giving control over the precision/overflow balance.

Spatial Reversal in IFFT Output

Challenge: The correlation response map computed via IFFT exhibited a 180° spatial reversal: the peak appeared in the wrong quadrant of the response image, causing the peak finder to report sign-inverted displacement vectors.

Measure: This is a known artefact of using the forward FFT to approximate the IFFT via conjugation — the output is circularly shifted by $N/2$ in both dimensions. The fix was to apply `numpy.roll` (circular shift by $N/2$) in the Python reference model and the equivalent register-shift in the RTL peak finder post-processing stage. The fix was verified simultaneously in simulation and hardware testbench.

LWIR Sensor Procurement

Challenge: LWIR thermal imaging sensors suitable for the application (e.g., FLIR Lepton 3.5, Boson 640) are ITAR-controlled and carry 12–16 week academic procurement lead times, far exceeding the MVP timeline.

Measure: An ESP32-S3 with OV2640 camera (already available in the lab) was deployed as a pin-compatible surrogate using identical UART frame-transfer. All hardware and software development has continued uninterrupted on this surrogate. Procurement of the LWIR sensor is in progress.

Strategic Challenges

Evaluator Pushback on Market Differentiation

Challenge: During the MVP panel review (27 February 2026), evaluators noted that triangulation methods and standard IR-guided seekers are already deployed commercially, questioning the project’s novelty.

Measure: The team clarified the LWIR differentiator: standard IR requires an active heat source (engine exhaust, propulsion plume), while LWIR detects intrinsic thermal signatures of thermally passive objects (structures, grounded vehicles, infrastructure). A refined competitive analysis was prepared and backup slides covering the four key target scenarios were developed (see Section 9).

GPS Reliability Framing

Challenge: Evaluators questioned whether a vision-based seeker adds value when GPS guidance is available.

Measure: The system was reframed as a fail-safe and redundancy layer for GPS-denied or GPS-jammed environments, not a GPS replacement. Four operational scenarios demonstrating necessity were formalised: target relocation, intermittent jamming, last-minute jamming, and multi-sensor redundancy.

Domain Expert Details

Mentor and Domain Expert	
Name:	Dr. Venkatnarayan Hariharan
Designation:	Faculty, Department of Electrical Engineering
Institution:	Shiv Nadar Institution of Eminence, Greater Noida
Email:	(Department of Electrical Engineering, SNU)
Expertise:	VLSI Design, FPGA-based Signal Processing, Embedded Systems, RISC-V Architecture
Role:	Technical mentor for RTL design, algorithm selection, AXI interconnect architecture, and hardware–software co-design.
PoC Contributions:	Supervised KCF/MOSSE RTL pipeline design; validated AXI4-Lite register map; reviewed Python–RTL co-simulation methodology; guided fixed-point arithmetic strategy and FPGA resource optimisation.

Dr. Hariharan’s background in VLSI and FPGA signal processing was critical in the decision to implement KCF fully in RTL (rather than software) and in structuring the AXI4-Lite interface for clean processor integration. His guidance on the iterative single-butterfly architecture reduced the FPGA resource footprint to a single multiplier set while maintaining a fully verified, timing-closed pipeline.

MVP Panel Feedback and Improvements Made

The MVP review was held on **27 February 2026**. The following sections document each piece of feedback and the specific improvements made in response.

Feedback 1: Existing Market Solutions

Evaluator Feedback

“Alternative triangulation methods and pre-existing tracking solutions are currently deployed in the market for this specific problem statement.”

Improvement: A structured competitive analysis was conducted distinguishing this system from existing solutions on two axes: (1) *hardware sovereignty* — the entire compute stack runs on SHAKTI RISC-V, eliminating reliance on foreign CPUs/GPUs and supply-chain vulnerabilities; (2) *thermal sensing modality* — the LWIR sensor enables tracking of thermally passive objects not detectable by standard IR or EO seekers. Dedicated slides addressing these differentiators were prepared for future panel evaluations.

Feedback 2: Standard IR vs. LWIR

Evaluator Feedback

“For tracking moving targets with active heat signatures (jet exhausts), standard IR is a highly common, easily deployable industry standard.”

Improvement: The team acknowledges this for hot-target scenarios. The value proposition was sharpened: the system’s primary differentiator is tracking *thermally passive* objects (stationary vehicles, structures, infrastructure) that emit negligible active thermal radiation but retain distinct LWIR signatures relative to their thermal background. This positions the system as complementary to standard IR, not a direct competitor.

Feedback 3: Human-in-the-Loop Alternative

Evaluator Feedback

“Exploring a controllable, human-in-the-loop guidance system was suggested as an alternative architecture.”

Improvement: The human-in-the-loop architecture was evaluated. While viable for loitering munitions with a stable datalink, it introduces >500 ms round-trip latency and is vulnerable to communication jamming in contested environments. The autonomous on-board architecture is retained as the primary approach; human-in-the-loop is acknowledged as a parallel variant applicable to different mission profiles.

Feedback 4: Role of GPS

Evaluator Feedback

“GPS must be treated strictly as a last resort rather than a primary navigation tool in specific tactical environments.”

Improvement: This feedback directly validates the project’s core premise. The narrative was refined to position the IIR seeker as the *primary* terminal-phase guidance mechanism, with GPS serving only midcourse navigation. Four target scenarios where the system is operationally necessary were formalised:

1. **Target Relocation:** GPS coordinates recorded at mission launch become stale if the target has moved. The IIR seeker performs live visual acquisition.
2. **Intermittent Jamming:** In remote environments where adversarial jammers toggle unpredictably, the seeker provides uninterrupted terminal guidance.
3. **Last-Minute Jamming:** When jammers are deployed at the final stages of an engagement as a countermeasure, the autonomous seeker serves as a critical fail-safe.
4. **System Redundancy:** Acts as a second independent sensor to validate, support, or override the primary navigation system.

Action Items from MVP Review

Action Item	Assignees	Status
Benchmark image processing pipeline latency empirically	Tushar & Akhil	✓ Done
Create backup slides for refined target scenarios	Kalrav & Akansh	✓ Done

The benchmarking task produced the timing characterisation in Section 5.2.2: 19,300 cycles at 100 MHz = 193 μ s per detection decision — well within the <5 ms budget required for real-time guidance at 10 FPS.

Way Forward

Immediate Priorities

1. **Complete continuous video DMA loop:** Implement double-buffered DMA and validate >10FPS end-to-end throughput from the ESP32-S3 surrogate through the FPGA pipeline to the RTOS `co-ord_q`.
2. **IMU integration:** Wire and validate UART/SPI communication with the IMU; implement the sensor fusion logic in the SHAKTI Control Logic RTOS task.
3. **Vivado place-and-route closure:** Complete post-implementation static timing analysis and generate the programming bitstream for physical FPGA testing.

4. **End-to-end hardware test:** Run the full pipeline on physical hardware — sensor → FPGA → AXI → SHAKTI RTOS → flight command output — and measure real-world closed-loop latency.
5. **LWIR sensor swap:** Finalise procurement and replace the ESP32-S3 with the LWIR thermal sensor. Validate that thermal frames produce correct tracking output with no algorithm changes.
6. **Scale to 64×64 patch:** Upgrade the FFT core from 32-point to 64-point to improve tracking accuracy at greater target distances and under partial occlusion.

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